

06_Implementation_Plan

Agentic Regulatory Governance & Unified Simulation

Team Rocket

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1. Scope & Strategy

Build a demo-credible MVP that runs the full pipeline (circular → obligations → mapping → stress test → RRI → replay → action plan → report) end-to-end on seeded data, prioritizing the **Regulatory Stress Test** and **RRI** as the differentiating features to polish first.

2. Phased Plan

Phase 0 — Setup (Day 0)

- Repo scaffolding: FastAPI backend, React+Tailwind+TS frontend, Docker Compose (Postgres, ChromaDB).
- Postgres schema migration from Backend Schema doc.
- LLM provider abstraction layer (single interface, config-driven provider selection).
- **Verify:** docker compose up boots all services; health check endpoints return 200.

Phase 1 — Ingestion & Extraction

- Circular upload endpoint + PDF/OCR parsing pipeline.
- Regulation Intelligence Agent: chunk circular → LLM structured extraction → obligations table.
- Frontend: Upload screen + Obligation Review table (edit/confirm).
- **Verify:** Upload a sample SEBI circular → obligations appear in review table within target time; manual edit persists correctly.

Phase 2 — Mapping

- Seed synthetic org policy corpus (5-10 policies/SOPs across 3-4 departments) and embed into vector store.
- Process Mapping Agent: retrieval + LLM relevance judgment → obligation_policy_mappings.
- Frontend: Mapping Review screen with confidence display and manual override.
- **Verify:** At least one obligation in the seed set correctly surfaces as “unmapped” (needed to demonstrate gap detection later).

Phase 3 — Stress Test & Findings (core differentiator — prioritize polish here)

- Stress Test Agent: implement the four check types (unimplemented / outdated / conflict / workflow gap) as a mix of deterministic rules (date comparisons, missing-mapping checks) and LLM reasoning (conflict/ambiguity detection).
- findings table population + severity assignment.
- Frontend: Findings view (grouped by severity/type).
- **Verify:** Seed data reproducibly generates a believable mix of findings (not zero, not overwhelming) — target 3-6 findings across severities for the demo dataset.

Phase 4 — Evidence Verification & RRI

- Evidence upload endpoint + document type tagging.
- Evidence Verification Agent: match evidence to obligations/findings, set status (Present/Stale/Missing).
- RRI Calculator: deterministic weighted scoring per formula in TRD; append-only `readiness_scores` writes.
- Frontend: Evidence upload UI, RRI Gauge component on dashboard with trend chart.
- **Verify:** Uploading evidence for a “Missing evidence” finding updates its status and the RRI recalculates and reflects the change live.

Phase 5 — Regulatory Replay & Advisor

- Build `regulatory_replays` generation: assemble the chain from existing relational data (circular → obligation → policy → workflow → gap → fix) plus LLM-generated narrative explanation.
- ARGUS Advisor: RAG-grounded chat endpoint scoped to org data.
- Frontend: Replay chain visual component (signature UI element) + persistent Advisor side panel.
- **Verify:** Every finding has a working Replay view; Advisor answers a scoped question (e.g., “why is evidence completeness low?”) using only org data, with no hallucinated obligations.

Phase 6 — Action Plan & Reporting

- Auto-generate `action_items` from open findings (priority = finding severity, deadline = heuristic offset from obligation deadline).
- Frontend: Action Plan table (editable owner/deadline/status).
- Report generation: compile RRI + findings + replays + action plan into a PDF (reportlab or server-side render).
- **Verify:** Full pipeline run produces a downloadable PDF report matching what’s shown on screen.

Phase 7 — Demo Hardening

- Seed a polished “before” dataset for live demo (moderate RRI ~70-86, 2-3 high-priority risks, at least one full Replay chain worth walking through).
- Smoke-test the exact 5-minute demo script end-to-end at least 3 times.
- Cut anything not needed for the demo script from the critical path (auth can be a stubbed single demo login; multi-tenant polish deferred).

3. Milestones & Verification Summary

Milestone	Verifiable Outcome
M1 — Ingestion works	Circular → obligations table populated and human-editable
M2 — Mapping works	Obligations linked to policies with visible confidence + unmapped flag
M3 — Stress test works	Findings generated with correct type/severity classification
M4 — RRI live	Score recalculates correctly on evidence/action changes
M5 — Replay + Advisor	Every finding has a coherent, traceable explanation chain
M6 — Action plan + report	End-to-end PDF export matches on-screen state
M7 — Demo-ready	Full 5-minute flow runs cleanly, repeatably, on seeded data

4. Risks & Mitigations

Risk	Mitigation
LLM extraction misses/misreads obligations	Human confirm gate before downstream agents run; use structured function-calling schema, not free text
Stress test produces implausible or zero findings on demo data	Curate seed policy corpus deliberately to include at least one

	of each gap type
LLM cost/latency during live demo	Pre-run the pipeline once before the demo and cache results; live-run only the specific step being narrated (e.g., evidence upload → RRI update)
Scope creep beyond hackathon window	Lock Phase 0-6 as MVP; anything else (multi-regulator, SSO, mobile) explicitly deferred per PRD non-goals

5. Post-Hackathon Roadmap (Not in MVP)

- Multi-regulator support (RBI, IRDAI) via pluggable regulation-source adapters.
- Automated document classification for evidence (reduce manual tagging).
- Continuous ingestion (webhook/scrapper for new SEBI circulars vs. manual upload).
- Fine-grained RBAC and multi-tenant hardening (row-level security, SSO).
- Benchmark suite for gap-detection recall against a labeled compliance dataset.